

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

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FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 2.6: AREA OF A PART OF A CIRCLE

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation document. Use it as a model to construct your own written answer to the driving question. Read each section carefully, study the exemplar responses, then write your own explanation in your exercise book or on the platform. Use clear mathematical language, include labelled formulae, and always connect your answer back to real-life situations like the Laikipia farmer's centre-pivot irrigation system. Write in full sentences unless a formula or calculation is shown.

Section 1: The Full Circle as Your Starting Point

Before you can find the area of any PART of a circle, you must be confident with the whole. Explain what the formula $A = \pi r^2$ means, identify each variable, and describe why it is the foundation for every other formula in this sub-strand.

The area of a complete circle is given by $A = \pi r^2$, where A is the area in square metres (or any square unit), $\pi \approx 3.14159$ is a mathematical constant, and r is the radius — the distance from the centre water source to the tip of the irrigation pipe. For the Laikipia farmer, if the centre-pivot pipe is 200 m long, the total watered area is $A = \pi \times 200^2 = \pi \times 40\,000 \approx 125\,664 \text{ m}^2$. This full-circle area is written on the farmer's title deed. Every formula for a ring, a sector, or a segment is derived by either subtracting one circle's area from another, or by taking a fraction of this full-circle formula. That is why $A = \pi r^2$ is always Step 1 in your working.

Connect your answer to the Laikipia farm.

Section 2: The Area of an Annulus (Ring)

The farmer wants to add a second, outer pipe ring to the irrigation system. The region between the inner pipe and the outer pipe is called an annulus (ring). Derive the formula for the area of an annulus, define every variable, and use the Laikipia farm to show a fully worked numerical example. Explain how this helps the farmer avoid waste.

An annulus is the region between two concentric circles (circles that share the same centre). If the inner circle has radius r and the outer circle has radius R , then:

$$\text{Area of annulus} = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$$

Where R is the outer radius and r is the inner radius, with $R > r$.

Worked example: The farmer's first pipe has radius $r = 200$ m. The new outer pipe will reach $R = 350$ m from the same central water source. The area of the new ring section that will be watered is:

$$\begin{aligned} A &= \pi(350^2 - 200^2) \\ &= \pi(122\,500 - 40\,000) \\ &= \pi \times 82\,500 \\ &\approx 259\,182 \text{ m}^2 \\ &\approx 25.9 \text{ hectares} \end{aligned}$$

This tells the farmer exactly how many additional hectares of maize can be planted in the new ring, so they can buy the right amount of seed and fertiliser without over-spending or wasting resources. The formula works because you are simply subtracting the area already covered by the inner circle from the total area covered by the outer circle.

Section 3: The Area of a Sector (Wedge-Shaped Slice)

When the centre-pivot pipe has only rotated part-way around the field, it sweeps out a wedge-shaped region called a sector. Derive the formula for the area of a sector, explain what the angle θ represents, and show a fully worked example. Include both the degree version and the radian version of the formula.

A sector is the 'pizza-slice' region bounded by two radii and the arc between them. The angle θ at the centre tells you what fraction of the full circle the sector occupies.

Because a full circle has 360° , a sector with angle θ° is a fraction $\theta/360$ of the whole circle:

$$\text{Area of sector (degrees)} = (\theta/360) \times \pi r^2$$

In radians (where a full circle = 2π radians):

$$\text{Area of sector (radians)} = \frac{1}{2}r^2\theta$$

Worked example (degrees): The Laikipia farmer's pipe ($r = 200$ m) has rotated through 135° by midday. How much of the field has been watered?

	$A = (135/360) \times \pi \times 200^2$ $= (3/8) \times \pi \times 40\,000$ $= (3/8) \times 125\,664$ $\approx 47\,124 \text{ m}^2$ $\approx 4.71 \text{ hectares}$ Worked example (radians): $135^\circ = 135 \times (\pi/180) = 3\pi/4$ radians. $A = \frac{1}{2} \times 200^2 \times (3\pi/4)$ $= \frac{1}{2} \times 40\,000 \times 2.356$ $\approx 47\,124 \text{ m}^2 \checkmark \text{ (same answer, confirming both formulae agree)}$ Knowing this area helps the farmer estimate how much water has been used and whether the water tank needs refilling before the pipe completes a full rotation.
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Section 4: The Area of a Segment	
A segment is the region between a chord and the arc it cuts off — like a slice with the pointed tip removed. Explain how you find the area of a segment by combining the sector formula and the triangle formula. Show a fully worked example and describe one real-life situation in Kenya where this calculation matters.	A segment is formed when a straight chord cuts across a circle. The region between the chord and the arc is called a minor segment (smaller part) or major segment (larger part). To find the area of a minor segment with central angle θ: Step 1 — Find the area of the sector with the same angle θ. Step 2 — Find the area of the triangle formed by the two radii and the chord. Step 3 — Subtract: Area of segment = Area of sector – Area of triangle Formula (angle in degrees): Area of segment = $(\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta$ Formula (angle in radians): Area of segment = $\frac{1}{2}r^2(\theta - \sin \theta)$ Worked example: A circular water-storage tank in Kisumu has radius $r = 3$ m. A flat concrete dividing wall (chord) cuts across it, creating a central angle of $\theta = 90^\circ$. Area of sector = $(90/360) \times \pi \times 3^2 = \frac{1}{4} \times \pi \times 9 \approx 7.07 \text{ m}^2$ Area of triangle = $\frac{1}{2} \times 3^2 \times \sin 90^\circ = \frac{1}{2} \times 9 \times 1 = 4.5 \text{ m}^2$ Area of segment = $7.07 - 4.5 = 2.57 \text{ m}^2$ Real-life connection: A Nairobi metalwork artisan cutting a decorative arch into a circular steel gate needs to know the area of the segment to calculate the mass of metal removed and the cost of the leftover scrap. Segment calculations prevent expensive errors.

Section 5: Answering the Driving Question — Putting It All Together

Now answer the driving question directly and completely: 'How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers, and artisans can make accurate plans and avoid waste?' In your answer, name all three sub-regions (annulus, sector, segment), state their formulae, and explain how choosing the correct formula depends on the shape of the region you are working with. Use the Laikipia farmer as your unifying example.

We can calculate the exact area of any part of a circle by identifying which sub-region we are dealing with and applying the correct derived formula.

1. Annulus (ring between two circles): $A = \pi(R^2 - r^2)$

→ Use this when two concentric pipes or circular boundaries create a ring-shaped region, as when the Laikipia farmer adds an outer irrigation ring.

2. Sector (wedge/slice): $A = (\theta/360)\pi r^2$ or $A = \frac{1}{2}r^2\theta$ (radians)

→ Use this when a pivot or boundary has rotated through a known angle, sweeping a wedge-shaped area, as when the irrigation pipe has completed only part of its rotation.

3. Segment (region between a chord and an arc): $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta$

→ Use this when a straight edge or wall cuts across the circular region, as when a path or fence cuts across a circular plot.

The key to avoiding waste is selecting the right formula before calculating. Confusing a sector with a segment, for example, would lead to an over-estimate of the watered area, causing the farmer to under-fill the water tank and leave part of the maize crop unwatered. By mastering all three formulae and understanding the geometry behind each, farmers can budget seeds and water accurately, engineers can design efficient structures, and artisans can cut materials precisely — eliminating waste and saving money across Kenya's industries.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Conceptual Understanding of Circle Parts	Accurately defines and distinguishes all three sub-regions (annulus, sector, segment) using precise mathematical language. Clearly explains how each is derived from the full-circle formula $A = \pi r^2$.	Correctly defines at least two of the three sub-regions and shows a sound understanding of how they relate to the full circle, with minor gaps in precision.	Shows partial understanding of one or two sub-regions. Definitions are incomplete or confused (e.g., sector and segment are mixed up). Limited connection to the full-circle formula.
Correct Use and Derivation of Formulae	States all three formulae correctly (annulus, sector in degrees AND radians, segment). Clearly derives or explains each formula step-by-step. No errors in formula notation.	States at least two formulae correctly and attempts derivation. May omit the radian form of the sector or make a minor notational error, but overall structure is mathematically sound.	Recalls one formula correctly or makes significant errors in formula notation (e.g., writes $A = \pi d^2$ or omits subtraction for the segment). Little or no attempt at derivation.
Accuracy of Numerical Worked Examples	All worked examples are fully shown step-by-step, answers are correct to at least 2 decimal places or significant	Most steps are shown and answers are correct. May make one arithmetic slip or omit units in one step, but the method is	Worked examples are incomplete or contain significant arithmetic errors. Steps are missing or out of order, making

	figures, and units (m ² , hectares, etc.) are stated consistently throughout.	clearly correct and the final answer is reasonable.	the method difficult to follow. Units are missing or wrong.
Connection to Real-Life Kenyan Contexts	Every formula and calculation is explicitly linked to a specific, realistic Kenyan context (e.g., Laikipia irrigation, Kisumu water tank, Nairobi artisan). Explanations clearly show how the mathematics prevents waste or supports decision-making.	At least two real-life Kenyan contexts are used meaningfully. The link between the mathematics and the practical outcome (avoiding waste, planning resources) is stated but may lack specific detail.	Real-life connection is vague, generic (e.g., 'used in real life'), or absent. The Kenyan context from the phenomenon is not referenced, or the connection does not demonstrate understanding of why the calculation matters.
Completeness and Clarity of the Driving Question Response	The final section directly and fully answers the driving question, naming all three sub-regions, their formulae, and the decision rule for choosing between them. Writing is clear, logical, and well-organised with no unexplained jumps.	The driving question is answered and covers most key points. At least two sub-regions are addressed. The response is mostly clear but may lack a decision rule or a unified concluding statement.	The driving question is only partially answered. One or more sub-regions are missing. The response is disorganised, too brief, or does not synthesise the formulae into a coherent answer.